

TrES-3b

R-2754

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150619 · created 2026-09-02T23:48:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457192.75395 +/- 0.00094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.0093
Transit depth (Rp/Rs)^2	3.84 +/- 0.36 [%]
Orbital Inclination (inc)	82.68 +/- 0.42
Airmass coefficient 1 (a1)	1.014 +/- 0.011
Airmass coefficient 2 (a2)	-0.012 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.04 %
Best Comparison Star	#2 - [549, 420]
Optimal Method	PSF photometry
Transit Duration (day)	0.0653 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.0093 vs 0.16309 (deviation 3.54σ)
Duration fit vs published	0.0653 d vs 0.0567 d (deviation 2.7σ)
Mid-time O-C	2.1 min
Depth SNR	21.1
Residual scatter	1.04 %

Light curve

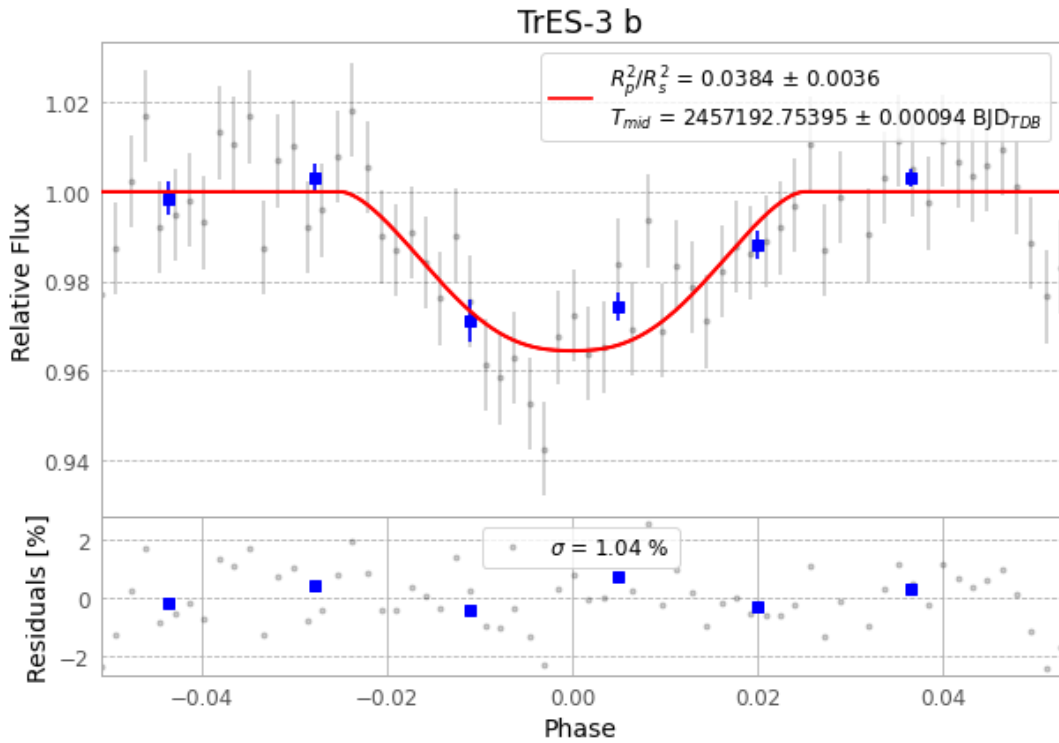


Figure 1 — FinalLightCurve_TrES-3 b_2015-06-18.png (SHA-256 fd174bba4b92c65c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 209], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b61a57e3bd348084...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3150619042412.FITS → TRES-3150619073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150619.log (baac6cd5774c...), FinalLightCurve_TrES-3 b_2015-06-18.png (fd174bba4b92...), FinalLightCurve_TrES-3 b_2015-06-18.csv (b84a8da42f46...)

Compute resources

Wall time 146.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:48Z.